Colégio **Agostiniano Mendel**

Componente Curricular: Física – Dinâmica
Professor(a): Rosana C. Pádua **Série:** **Data:** Março/2026
Conteúdo: Trabalho de uma Força - Exercícios

1\. a) $\tau_F = F.d.\cos\theta$

$\tau_F = 20.3.\frac{\sqrt{3}}{2} \Rightarrow \tau_F = 30\sqrt{3}\,J$

b) $\tau_F = F.d.\cos 120°$

$\tau_F = 40.4.(-0{,}5) \Rightarrow \tau_F = -80\,J$

2\. * $F_{2x} = F_2.\cos\theta$

$F_{2x} = 100.0{,}6 = 60\,N$

* $F_{2x} > F_4$ (há deslocamento horizontal p/a direita)

Obs: Vertical $\Rightarrow$ • $F_{2y} = F_2.\text{sen}\theta \Rightarrow F_{2y} = 100.0{,}8 = 80\,N$

• $F_2 + F_{2y} = 120 + 80 = 200\,N$ ∴ Não há deslocamento na vertical, pois F_3 e $F_{2y} = F_1$

* $\tau_{F3} = 0$ e $\tau_{F1} = 0$ (são ⊥ ao deslocamento)

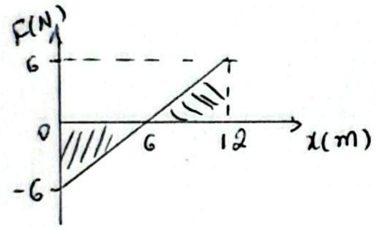
• $\tau_{F2} = F_2.d.\cos\theta \Rightarrow \tau_{F2} = 100.5.0{,}6$

$\tau_{F2} = 300\,J$

• $\tau_{F4} = -F_4.d \Rightarrow \tau_{F4} = -50.5$

$\tau_{F4} = -250\,J$

3\.



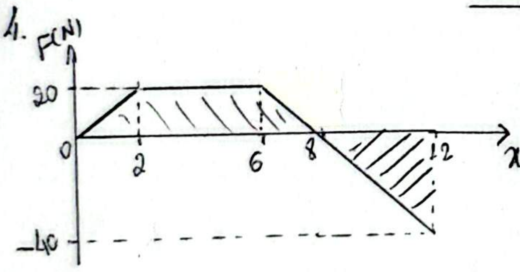
a) • 0 a 6m $\Rightarrow \tau_1 = -A \Rightarrow \tau_1 = -\frac{6.6}{2} \Rightarrow \tau_1 = -18\,J$

b) • 6m a 12m $\Rightarrow \tau_2 = +A \Rightarrow \tau_2 = \frac{6.6}{2} \Rightarrow \tau_2 = 18\,J$

c) • 0 a 6m $\Rightarrow \tau = \tau_1 + \tau_2$

$\tau = -18 + 18 \Rightarrow \tau = 0$

4\.



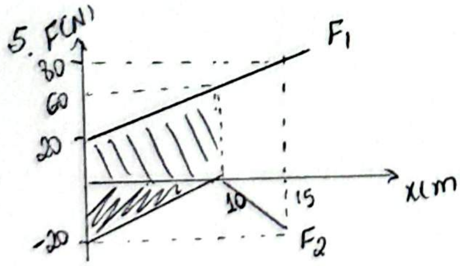
a) • 0 a 8m $\Rightarrow \tau = A \Rightarrow \tau_1 = \left(\frac{8+4}{2}\right).20 \Rightarrow \tau = 120\,J$

b) 8m a 12m $\Rightarrow \tau_2 = -\frac{4.40}{2} \Rightarrow \tau_2 = -80\,J$

c) 0 a 12m $\Rightarrow \tau = \tau_1 + \tau_2$

$\tau = 120 + (-80) \Rightarrow \tau = 40\,J$

5\.



a) $\tau_{F1} = +A$

$\tau_{F1} = \left(\frac{60+20}{2}\right).10$

$\tau_{F1} = 400\,J$

b) $\tau_{F2} = -A$

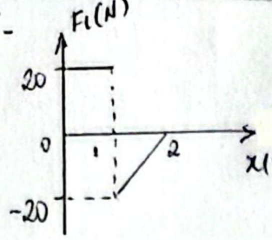
$\tau_{F2} = -\frac{10.20}{2}$

$\tau_{F2} = -100\,J$

c) $\tau_R = \tau_1 + \tau_2$

$\tau_R = 400 + (-100) \Rightarrow \tau_R = 300\,J$

6-



a) $\cdot\ \tau_{F2} = -F_2 \cdot d$

$\tau_{F2} = -5 \cdot 2$

$\tau_{F2} = -10J$

b) $\tau_{F1} = \tau_1 + \tau_2$

$\tau_{F1} = 20 \cdot 1 + \left(-\frac{1 \cdot 20}{2}\right)$

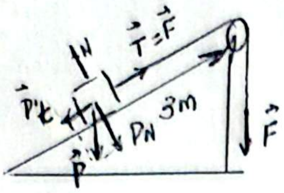
$\tau_{F1} = 20 - 10 \Rightarrow \tau_{F1} = 10J$

c) $\tau_R = \tau_1 + \tau_2$

$\tau_R = 10 + (-10)$

$\tau_R = 0$

7.



$*$ Fio ideal $\Rightarrow F = T$ $\quad *$ MU $\Rightarrow F_R = 0 \Rightarrow T = P_t$ $(P_t = 200J)$

a) $* \ \tau_F = F \cdot d$

$\tau_F = F \cdot \text{sen}\theta \cdot d$

$\tau_F = 4 \cdot 10^2 \cdot \frac{1}{2} \cdot 3$

$\tau_F = 600J$

b) $\tau_P = \tau_{Pt}$

$\tau_P = -P_t \cdot d$

$\tau_P = -200 \cdot 3$

$\tau_P = -600J$

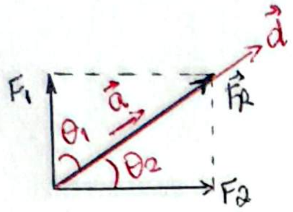
$(*$ Opção $\tau_P = -P \cdot h$ $\quad *\ \text{sen}30° = \frac{h}{d}$

$\tau_P = -4 \cdot 10^2 \cdot 1{,}5$ $\quad \frac{1}{2} = \frac{h}{3}$

$\tau_P = -600J$ $\quad h = 1{,}5m$

$*$ Peso é força conservativa

8.



a) $F_R = 10N$

b) $\cdot\ F_R = m \cdot a$

$10 = 20 \cdot a$

$a = 0{,}5 m/s^2$

$\cdot\ \Delta S = \frac{a}{2}t^2$

$\Delta S = \frac{0{,}5}{2} \cdot 4^2 \Rightarrow \Delta S = 4m$

$d = |\Delta S| \Rightarrow d = 4m$

c) $\cdot\ \tau_R = F_R \cdot d$

$\tau_R = 10 \cdot 4 \Rightarrow \tau_R = 40J$

$\cdot\ \tau_{F1} = F_1 \cdot d \cos\theta_1$

$\tau_{F1} = 6 \cdot 4 \cdot \frac{6}{10} \Rightarrow \tau_1 = 14{,}4J$

$\cdot\ \tau_{F2} = F_2 \cdot d \cos\theta_2$

$\tau_{F2} = 8 \cdot 4 \cdot \frac{8}{10} \Rightarrow \tau_2 = 25{,}6J$

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9. $\cdot\ \tau_P = 0$

b) $\cdot\ \tau_N = 0$

a) $\tau_F = F \cdot d \cdot \cos 45°$

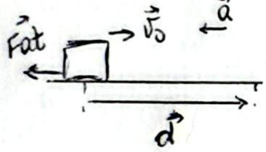
$\tau_F = 18\sqrt{2} \cdot 0{,}3 \cdot \frac{\sqrt{2}}{2} \Rightarrow \tau_F = 5{,}4J$

c) $\cdot\ \tau_{Fat} = -Fat \cdot d$

$\tau_{Fat} = -12 \cdot 0{,}3 \Rightarrow \tau_{Fat} = -3{,}6J$

d) $\tau_R = \tau_P + \tau_N + \tau_F + \tau_{Fat} \Rightarrow \tau_R = 5{,}4 + (-3{,}6) \Rightarrow \tau_R = 1{,}8J$

10.



$\cdot\ v^2 = v_0^2 + 2ad$

$0 = 8^2 + 2 \cdot a \cdot d$

$-2ad = 64$

$a = -\frac{32}{d}$

$\cdot\ F_R = m \cdot a$

$Fat = ma$

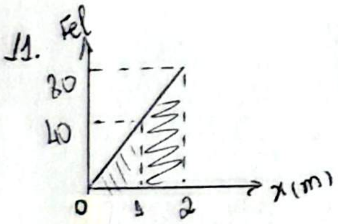
$Fat = 1 \cdot \frac{32}{d} \Rightarrow Fat = \frac{32}{d}$

$*\ \tau_{Fat} = -Fat \cdot d$

$\tau_{Fat} = -\frac{32}{d} \cdot d$

$\tau_{Fat} = -32J$

11.



$*\ \tau = -A$

Mola deformada

a) $\tau = -\frac{1 \cdot 40}{2}$

$\tau = -20J$

c) $\tau = -\left(\frac{80+40}{2}\right) \cdot 1$

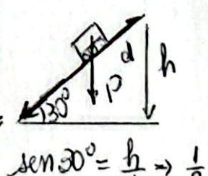
$\tau = -60J$

a) $Fel = kx$

$80 = k \cdot 2 \Rightarrow k = 40N/m$

12. $\Rightarrow m = 4t = 4 \cdot 10^3 kg$

$*\ 1kJ = 10^3 J$



$\text{sen}30° = \frac{h}{d} \Rightarrow \frac{1}{2} = \frac{h}{12} \Rightarrow h = 6m$

$\cdot\ \tau_P = P \cdot h \Rightarrow \tau_P = 4 \cdot 10^4 \cdot 6$

$\tau_P = 240kJ$

$*\ \Delta S = \frac{a}{2}t^2 \Rightarrow 12 = \frac{a}{2} \cdot 4^2$

$a = 1{,}5 m/s^2$

$*\ F_R = m \cdot a$

$F_R = 4 \cdot 10^3 \cdot 1{,}5 = 6 \cdot 10^3 N$

$\cdot\ \tau_R = F_R \cdot d$

$\tau_R = 6 \cdot 10^3 \cdot 12$

$\tau_R = 72kJ$